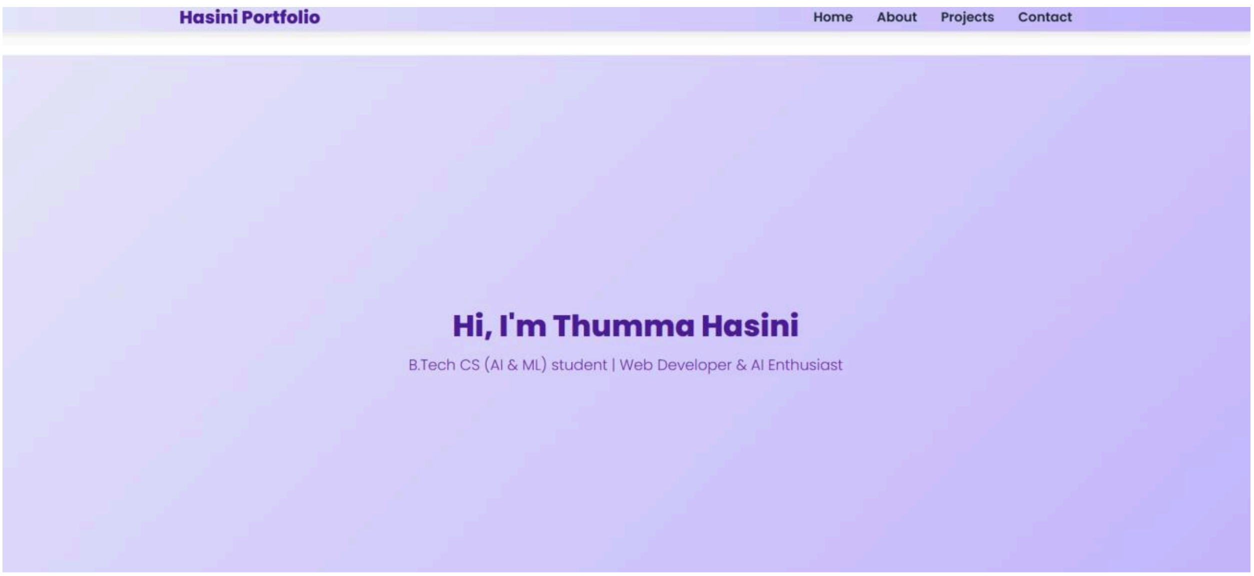
AI Assisted Coding Assignment - 14

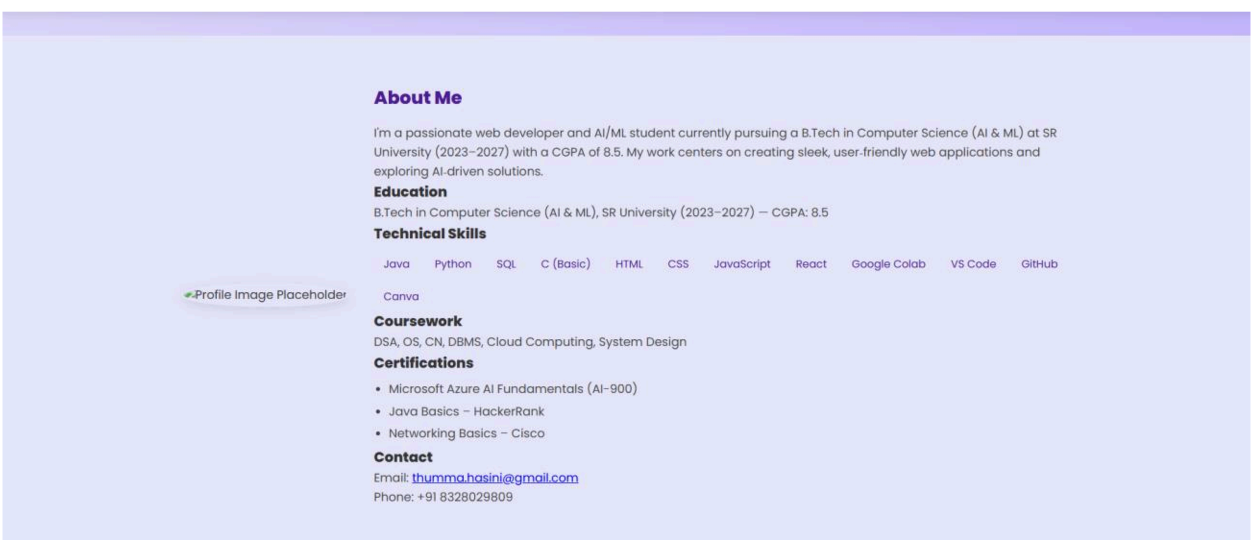
Chilukala Sai Architha(2303A52220) - Batch 37

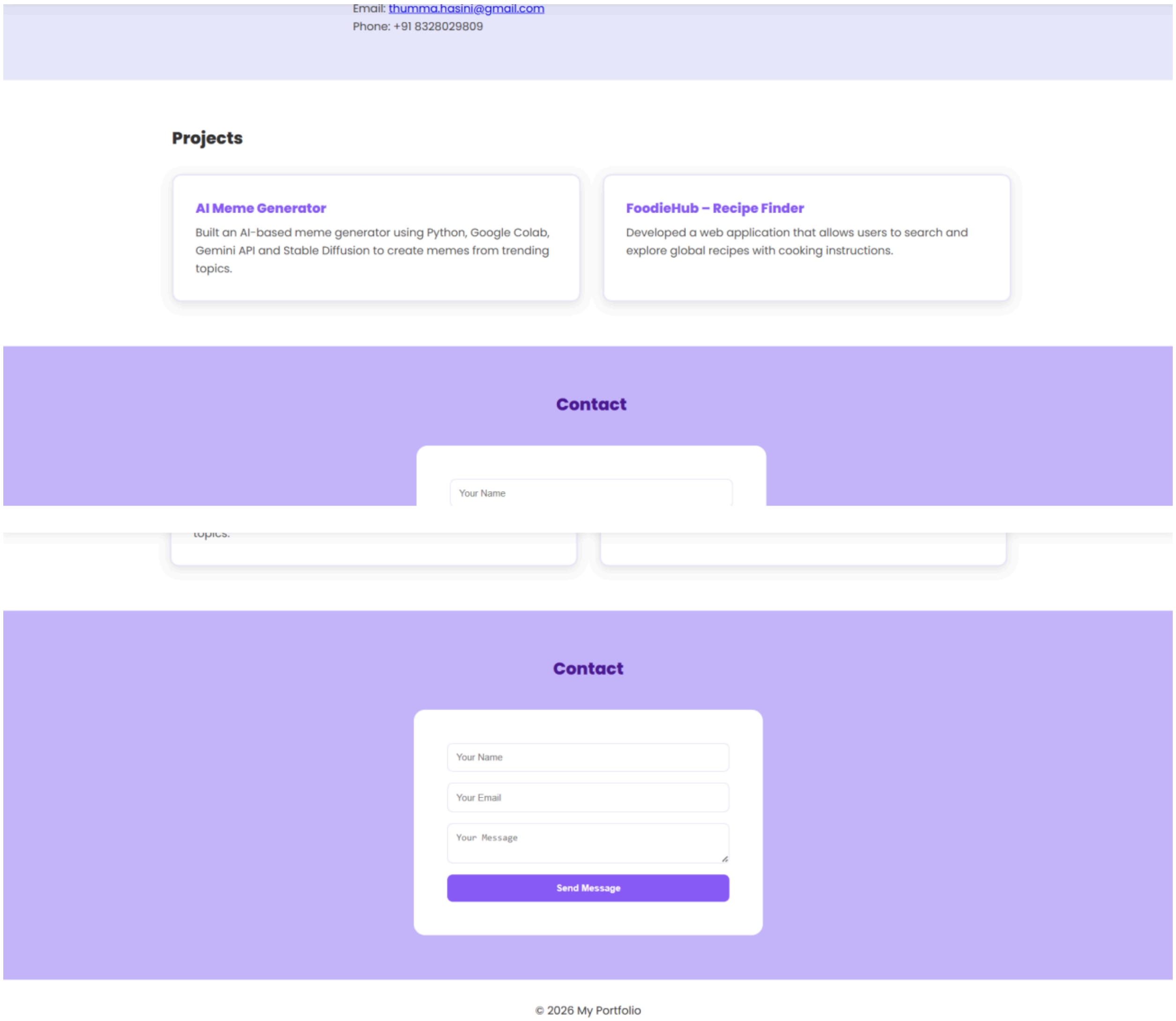
Task 1: AI-Assisted Personal Portfolio Website Scenario A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization. **Tasks** 1. Use GitHub Copilot to generate a semantic HTML structure with **sections:** o Home o About o Projects o Contact 2. Ask Copilot to suggest responsive CSS using Flexbox or Grid. 3. Customize AI-generated CSS to add: o Hover effects on project cards o Smooth scrolling navigation **Expected Outcome** •Clean and well-structured portfolio website •Responsive layout for desktop and mobile screens •Interactive project cards

Prompt Used in GitHub Copilot

Generate a responsive personal portfolio website using HTML, CSS, and JavaScript.
Create sections for Home, About, Projects, and Contact.
Use a modern white and lavender color palette.
Design a clean navigation bar with simple text links and hover effects.
Add a hero section with a heading and subtitle, an About section with image and text layout, project cards in a grid layout, and a centered contact form.
Ensure the layout is responsive and visually appealing.







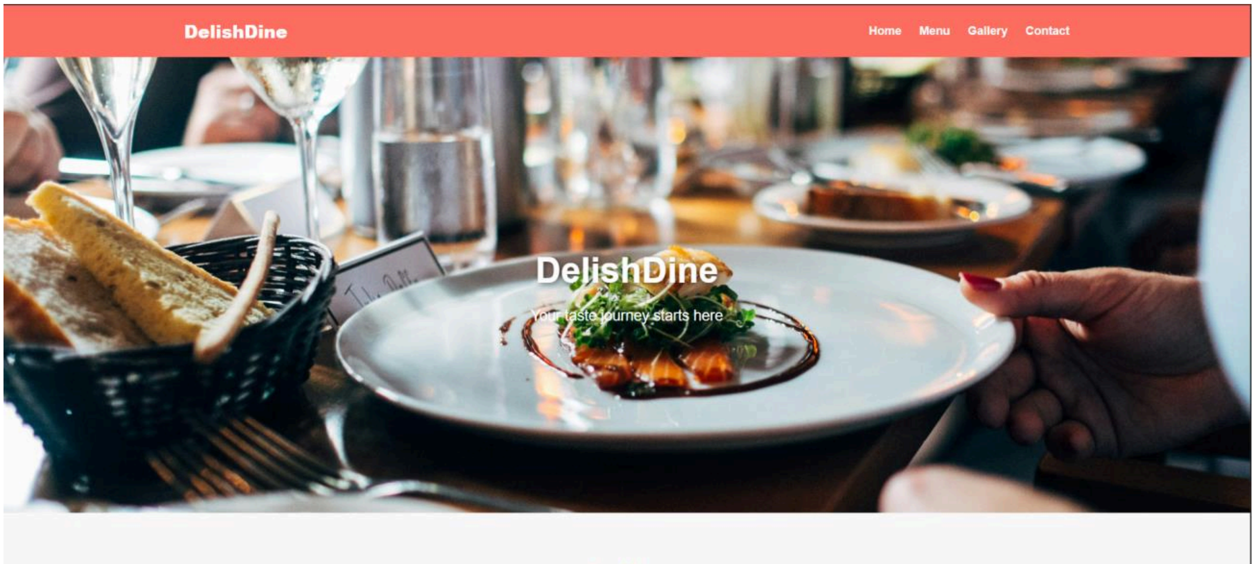
Explanation: GitHub Copilot was used to generate the basic HTML structure and CSS styling for the portfolio website. The design was customized with a white and lavender color palette and improved navigation styling. The final webpage includes responsive sections for Home, About, Projects, and Contact.

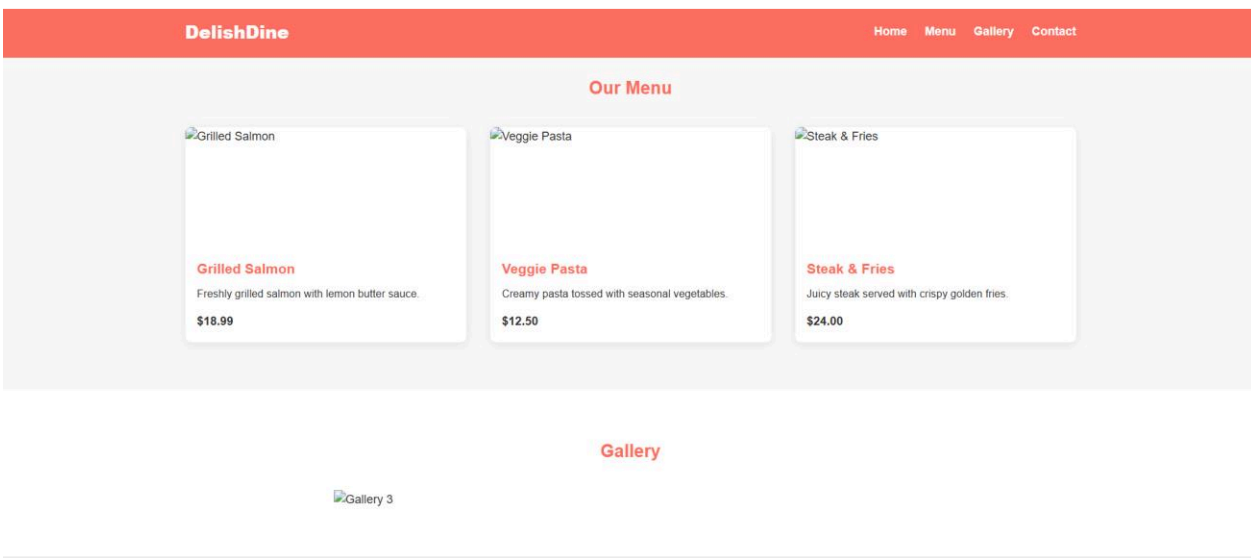
Task 2: AI-Generated Restaurant Landing Page

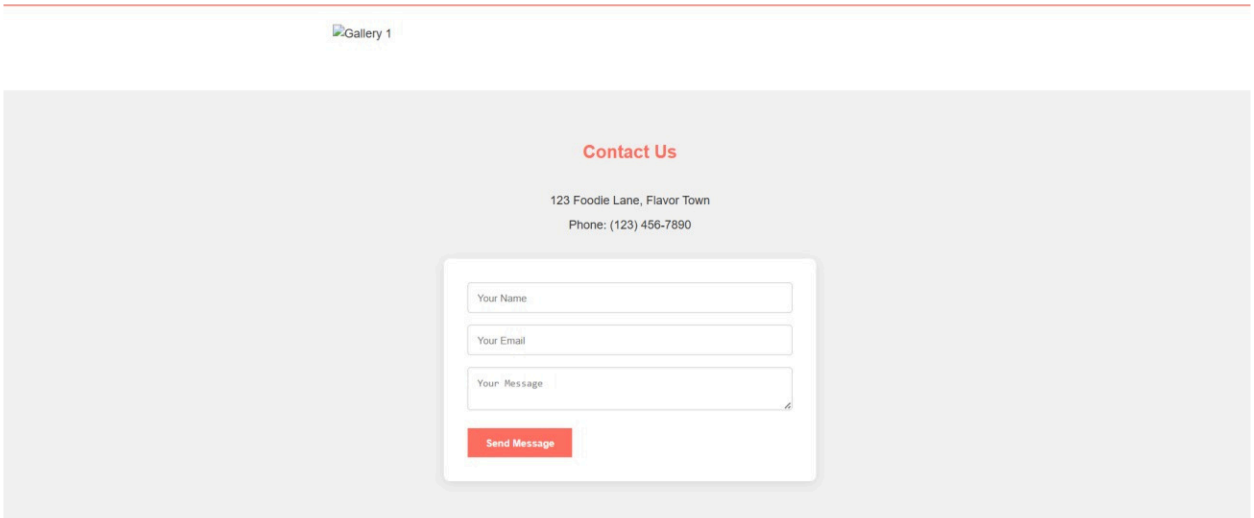
Scenario

A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components. **Tasks** 1. Use Copilot to create a navigation bar with links: o Home o Menu o Gallery o Contact 2. Generate a menu section styled with CSS cards. 3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions. 4. Customize animations or transitions for better user experience. **Expected Outcome** • Fully functional restaurant landing page • Responsive navigation bar • Working image slider with smooth transitions **Prompt Used in GitHub Copilot** Create a responsive restaurant landing page using HTML, CSS, and JavaScript.

Include a navigation bar with links: Home, Menu, Gallery, and Contact.
Add a hero section with a restaurant title and tagline.
Design the Menu section using card layouts that display food images, dish names, descriptions, and prices.
Create a Gallery section with a JavaScript image slider to show restaurant and food images.
Add a Contact section with address, phone number, and a simple contact form.
Ensure the layout is responsive and visually appealing.


DelishDine
Home
Menu
Gallery
Contact
DelishDine
Your taste journey starts here


DelishDine
Home
Menu
Gallery
Contact
Our Menu
Grilled Salmon
Grilled Salmon
Freshly grilled salmon with lemon butter sauce.
$18.99
Veggie Pasta
Veggie Pasta
Creamy pasta tossed with seasonal vegetables.
$12.50
Steak & Fries
Steak & Fries
Juicy steak served with crispy golden fries.
$24.00
Gallery
Gallery 3



Explanation: GitHub Copilot was used to generate the HTML structure and CSS styling for a restaurant landing page. The menu items were displayed using card layouts with food images and prices.A JavaScript image slider was implemented for the gallery section and the page was made responsive for different screen sizes.

Task 3: AI-Powered Event Registration Form

Scenario

SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input.

Tasks

1. Ask Copilot to generate an HTML form with the following fields:

o Name

o Email

o Phone Number

o Department

o Event Selection

2. Use AI assistance to style the form using CSS for readability and accessibility.

3. Implement JavaScript validation using Copilot:

o Email format validation

o Phone number length validation

o Required field checks

Expected Outcome

• User-friendly and visually appealing form

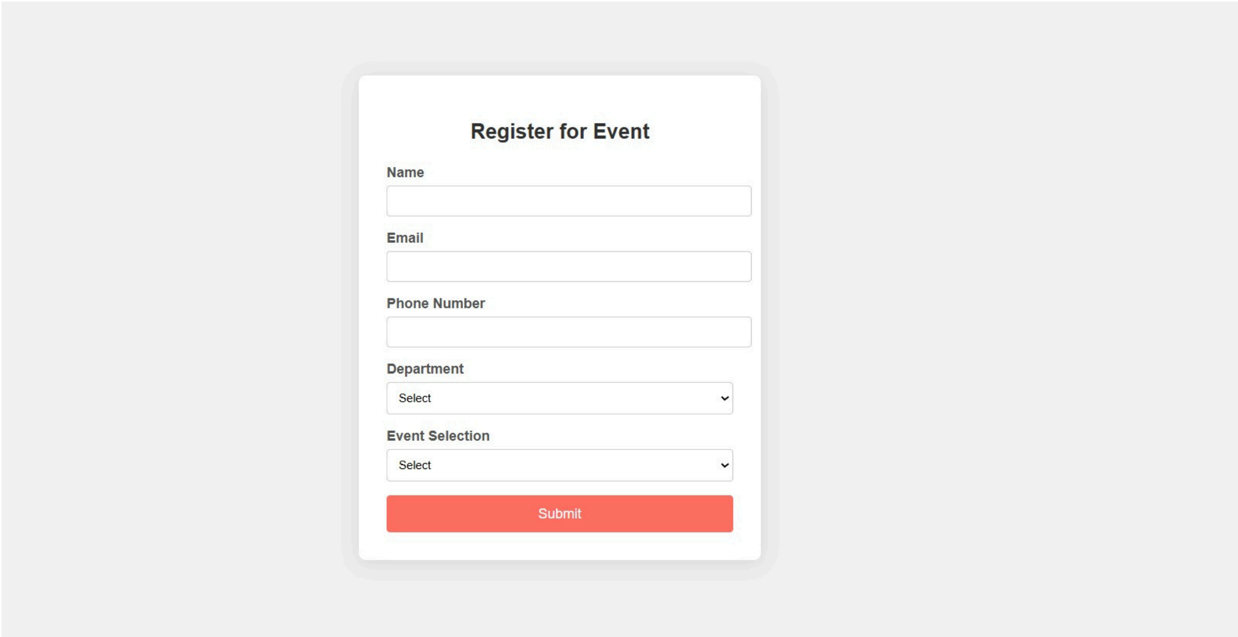
• Real-time validation messages

• Prevention of invalid submissions

Prompt Used in GitHub Copilot

Generate an event registration form using HTML, CSS, and JavaScript.

The form should include the following fields:
Name, Email, Phone Number, Department, and Event Selection.Style the form using CSS so that it is clean, centered, and easy to read.
Add JavaScript validation to check the email format, ensure the phone number has the correct length, and make all fields required before submission.
Display error message if the user enters invalid data.



Explanation: GitHub Copilot was used to generate a structured event registration form using HTML and CSS. JavaScript validation was added to check the email format, phone number length, and required fields. This ensures that users enter correct information before submitting the form.

Task 4: AI-Assisted E-Commerce Product Page

Scenario

A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality.

Tasks

1. Use Copilot to generate a grid-based product catalog using HTML and CSS.

2. Implement an “Add to Cart” feature using JavaScript.

3. Modify AI-generated code to include:

o Cart item counter at the top-right corner

o Visual feedback when a product is added

Expected Outcome

• Functional product catalog

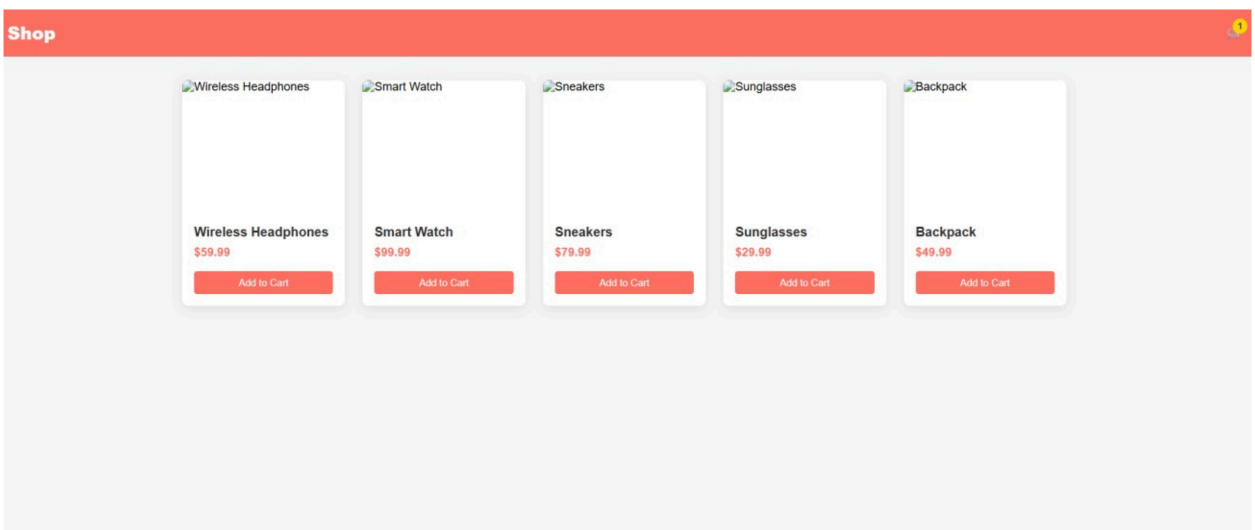
• Working cart counter

• Interactive user experience

Prompt Used in GitHub Copilot

Create a simple e-commerce product page using HTML, CSS, and JavaScript.

Display products in a responsive grid layout with product images, names, prices, and “Add to Cart” buttons.
Implement an Add to Cart feature using JavaScript so that when a user clicks the button, the cart counter at the top right increases.
Add visual feedback such as a small message or animation when a product is added to the cart.
Ensure the layout is responsive and visually organized.



Explanation: GitHub Copilot was used to generate an e-commerce product page with a grid-
based layout using HTML and CSS.JavaScript was implemented to create an “Add to Cart” functionality that updates the cart counter dynamically.Visual feedback was added to improve the user interaction when a product is added to the cart.

Task 5: Create a Responsive Web Page Layout

Instructions:

• Design a basic web page layout with a header, main content area, and footer using HTML and CSS.

• Use AI to assist in generating responsive CSS for different screen sizes.

• Ensure the layout is clean and visually organized.

Expected Output:

• A responsive web page with:

• Header with navigation links

• Main section with placeholder text/images

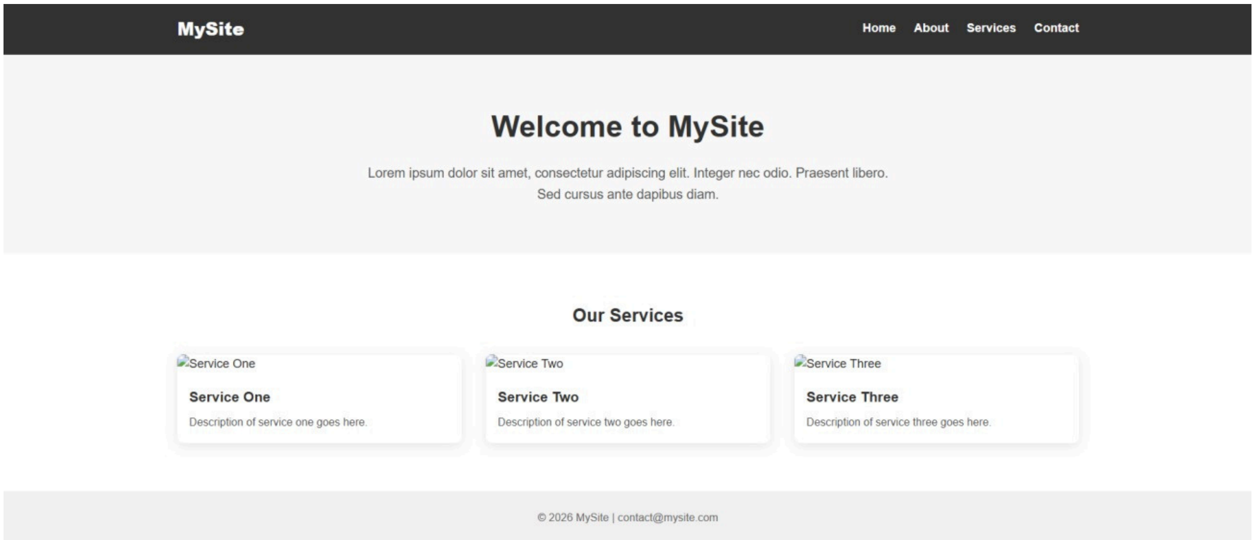
• Footer with copyright or contact info

• Layout adapts correctly to desktop, tablet, and mobile screen sizes.

Prompt Used in GitHub Copilot

Generate a responsive web page layout using HTML and CSS.

Create a header with navigation links, a main content section with placeholder text or images, and a footer with copyright or contact information.
Use responsive CSS techniques so the layout adjusts properly for desktop, tablet, and mobile screen sizes.
Ensure the design is clean, well-organized, and visually balanced.



Explanation: GitHub Copilot was used to generate a responsive web page layout using HTML and CSS. The page includes a header with navigation links, a main content section, and a footer. Responsive CSS techniques were applied to ensure the layout adapts correctly to different screen sizes.